

Executive Business Report

Retail Demand Forecasting & Inventory Intelligence System

Prepared For: Head of Supply Chain & CFO

Prepared By: Vivek Sagar Arora

Live Dashboard:

<https://retail-demand-forecasting-owmkr58fwkszreq3xasnwf.streamlit.app/>

Executive Summary

This project developed a retail demand forecasting system to support inventory planning and business decision-making. Historical sales data was analyzed to identify sales trends, seasonality, unusual sales events, and product demand patterns. Three forecasting models (SARIMA, Prophet, and XGBoost) were evaluated, with XGBoost providing the highest forecasting accuracy. An interactive Streamlit dashboard was developed to visualize forecasts, anomalies, and product demand segments, enabling data-driven inventory decisions.

Key Findings

- Office Supplies and Technology recorded the highest forecasted sales.
- Sales showed strong seasonal peaks during the year-end holiday period.
- The East region is expected to outperform the West region.
- XGBoost achieved the best forecasting performance (MAE: 14,763.81, RMSE: 18,337.41, MAPE: 14.48%).
- Demand varies across product categories, requiring category-specific inventory strategies.

Three-Month Sales Forecast

The XGBoost model predicts continued demand growth over the next three months. Office Supplies is expected to generate the highest sales, followed by Technology. The East region is projected to perform better than the West region. These forecasts can help optimize purchasing decisions and reduce stock shortages.

Top Anomalies Detected

Period	Likely Cause
March 2015	Seasonal promotional campaign
November 2018	Black Friday and festive sales
December 2018	Holiday shopping demand

Both Isolation Forest and Z-Score methods identified similar unusual sales periods, increasing confidence in the detected anomalies.

Product Demand Segmentation

Demand Segment	Recommended Strategy
High Value Products	Maintain controlled inventory to reduce holding costs.
Stable & Growing Demand	Maintain regular replenishment and steady stock levels.
High Volume, High Volatility	Maintain higher safety stock during peak demand periods.
Rapidly Growing Niche Products	Increase inventory gradually while monitoring demand trends.

Business Recommendations

1. Increase inventory for Office Supplies and Technology based on forecasted demand.
 2. Build additional stock before festive seasons to accommodate recurring sales spikes.
 3. Implement demand-based inventory policies instead of using a single stocking strategy for all products.
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Risk / Limitation

The forecasting model is based on historical sales data and does not consider external factors such as promotions, economic conditions, competitor activities, or supply chain disruptions. Incorporating these variables could further improve forecasting accuracy.

Conclusion

The Retail Demand Forecasting System combines forecasting, anomaly detection, demand segmentation, and an interactive dashboard to support inventory planning and business decision-making. It enables businesses to anticipate demand, optimize inventory, and improve supply chain efficiency through data-driven insights.